



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

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Re-Accredited by NAAC with 'A++' Grade

ASSIGNMENT No: 11

TITLE: Write a Java program to demonstrate the implementation of interfaces.

Problem Statement

Write a Java program to demonstrate the implementation of interfaces.

Learning Objectives

To implement interfaces for achieving abstraction and multiple inheritance of type.

Theory

An interface specifies a set of abstract methods that implementing classes must define. It provides complete abstraction and supports multiple inheritance of type. Interfaces improve modularity, loose coupling, and software flexibility. They allow different classes to implement common behavior while maintaining independent implementations. Interfaces are widely used in enterprise application development.

Conclusion:

The Java program was successfully implemented to demonstrate the use of interfaces. Interfaces were used to achieve abstraction by defining common methods that were implemented differently by multiple classes. The Printable and Switchable examples showed how interfaces support code reusability, loose coupling, and multiple inheritance of type. This experiment enhanced the understanding of object-oriented programming concepts and the practical use of interfaces in Java.

Exercise

1. Create an interface Printable and implement it in Student and Employee classes.

Code:

```
java > PrintableDemo.java > PrintableDemo
1  interface Printable {
2      void print();
3  }
4
5  class Student implements Printable {
6
7      public void print() {
8          System.out.println(x: "Student Details Displayed");
9      }
10 }
11
12 class Employee implements Printable {
13
14     public void print() {
15         System.out.println(x: "Employee Details Displayed");
16     }
17 }
18
19 public class PrintableDemo {
20     Run | Debug
21     public static void main(String[] args) {
22
23         Student s = new Student();
24         Employee e = new Employee();
25
26         s.print();
27         e.print();
28     }
29 }
```

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\cl404_30\Desktop\pratice> & 'C:\Program Files\Java\jdk1.8.0_25\bin\'
ratice_26094300\bin' 'PrintableDemo'
Student Details Displayed
Employee Details Displayed
PS C:\Users\cl404_30\Desktop\pratice>
```

2. Create an interface Switchable with a method turnOn(). Implement

the interface in Light and Fan classes to display the device status.

Code:

```
java > J SwitchableDemo.java > SwitchableDemo
1  interface Switchable {
2      void turnOn();
3  }
4
5  class Light implements Switchable {
6
7      public void turnOn() {
8          System.out.println(x: "Light is ON");
9      }
10 }
11
12 class Fan implements Switchable {
13
14     public void turnOn() {
15         System.out.println(x: "Fan is ON");
16     }
17 }
18
19 public class SwitchableDemo {
20     Run | Debug
21     public static void main(String[] args) {
22
23         Light l = new Light();
24         Fan f = new Fan();
25
26         l.turnOn();
27         f.turnOn();
28     }
29 }
```

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\cl404_30\Desktop\pratice> & 'C:\Program Files\Java\jdk1.8.0
ratice_26094300\bin' 'SwitchableDemo'
Light is ON
Fan is ON
PS C:\Users\cl404_30\Desktop\pratice>
```